Parth Samir Dalvi

Atlanta, GA | Relocation Ready | Remote Ready | 260-249-2495 | parthdalvi.atl@gmail.com | linkedin.com/in/parthdalvi04 | Portfolio | Github

Education

Master of Science in Computer Science - Software Architecture & Systems August 2024 – May 2026
Purdue University | Indiana, USA

Bachelor of Technology in Computer Science - Engineering August 2020 – July 2024
University of Pune | Maharashtra, India

Work Experience

Application Developer May 2025 – May 2026
Gameplay Inc | California, United States
Full-Stack Architecture | Python | Java | C++ | PostgreSQL | Microservices | Docker

- Architected scalable backend microservices, event-driven pipelines, and modular RESTful APIs in **Python and Java**, sustaining system throughput with **sub-20ms latency**.
- Engineered distributed data processing workflows handling asynchronous queue messaging, idempotency checks, and dead-letter queues across 10M+ daily transactions.
- Optimized relational database performance across **PostgreSQL and MySQL** by refactoring complex join queries and designing compound indexes, cutting read latency by **40%**.
- Built interactive, component-driven user interfaces in **TypeScript, React, and Next.js**, integrating real-time telemetry dashboards and reducing client-side load times by **25%**.
- Spearheaded automated testing and CI/CD deployment automation using **Docker and GitHub Actions**, sustaining a **95%** test coverage baseline in **Pytest and JUnit**.

Data Analyst March 2022 – July 2024
Weagile Software Solutions | Maharashtra, India
Data Pipelines | Python | SQL Relational Modeling | Telemetry Analytics | Git SCM

- Constructed automated ETL data extraction pipelines and analytics workflows in **Python and SQL**, reducing operational reporting turnaround by **38%**.
- Structured normalized schemas across enterprise databases, establishing automated validation triggers, boundary checks, and audit trails to guarantee zero data loss.
- Collaborated in Agile SDLC sprints alongside systems and QA engineers, performing root-cause analysis on data pipeline bottlenecks and reducing regression escapes by **42%**.
- Authored detailed technical architecture specifications, API documentation, and deployment runbooks in **Git** and MS Office.

Technical Skills

Languages & Systems Programming: Python - Expert, Java, C++, TypeScript, JavaScript ES6+, SQL - Advanced, Go Basics, Bash
Backend & Distributed Architectures: Microservices, RESTful APIs, FastAPI, Spring Boot, Node.js, GraphQL, WebSockets, Kafka, Redis
Frontend & Web Frameworks: React, Next.js, HTML5, CSS3, Tailwind CSS, Responsive Web Design, Component Architecture
Databases & Cloud Infrastructure: PostgreSQL, MySQL, MongoDB, AWS - S3, ECS, Microsoft Azure, Google Cloud, Docker, Kubernetes Basics
AI, Data Engineering & DevOps: Agentic AI Workflows, LLM APIs, Vector Embeddings, PySpark, Airflow, CI/CD, Git, Pytest, JUnit

Projects

Fibucks - Enterprise Financial Ledger Web Platform | **fibucks.com** Oct. 2025 – Jan. 2026
TypeScript, Next.js, Node.js, Python, PostgreSQL, OAuth 2.0, Webhooks, REST APIs

- Architected a full-stack personal finance application with **sub-15ms parsing latency**, supporting real-time transaction tracking and analytics.
- Constructed secure webhook receiver pipelines enforcing cryptographic signature verification, session validation, and idempotent data writes.
- Structured relational data models in PostgreSQL with foreign-key isolation and audit logging, eliminating concurrency collisions.
- Achieved **96% test coverage** across transactional workers, validation endpoints, and API adapters using automated test suites.

AgentPR - Autonomous AI Agent Engine Workflow Orchestration Platform Nov. 2025 – Feb. 2026
TypeScript, Python, LangChain, Multi-Step Tool Calling, AST Guardrails, Automated Triage, Pytest

- Constructed an autonomous multi-agent developer system in **Python and TypeScript** executing complex code analysis workflows.
- Designed multi-step tool-calling routines and evaluation feedback loops to analyze system states, diagnose bugs, and synthesize fixes.
- Engineered AST validation guardrails and automated safety checks, intercepting runtime regressions before production rollout.
- Maintained **96% test coverage** across task orchestration modules, API connectors, and execution harnesses using Pytest.

Autonomous Swarm Coordination Edge Telemetry Platform July 2025 – Dec. 2025
C++, Python, ROS, OpenCV, Raspberry Pi, Lightweight Messaging, Embedded Sockets

- Architected edge communication and spatial coordination software in **C++ and Python** deployed across distributed UAV hardware nodes.
- Engineered lightweight peer-to-peer socket messaging protocols, reducing inter-node communication latency by **20%** under heavy interference.
- Integrated computer vision pipelines for real-time object tracking and collision avoidance operating under strict hardware bounds.
- Commended with a formal Appreciation Letter from an Indian Air Force Squadron Leader for mission-critical software reliability.